Balaji Leninrajan

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Education

University of Waterloo 2023 – 2028
Bachelor of Software Engineering *Waterloo, ON, Canada*
Coursework: Operating Systems, Compilers, Computer Networks, Distributed Systems, Databases, FPGA Design

Experience

Finta | Software Engineering Intern | YC W23 *May 2026 – August 2026*
- **Reduced core product load time by over 98%**, from approximately 2 minutes to under 2 seconds, by replacing request-time aggregation with indexed PostgreSQL read models and rebuilding the interface in React
- Built a collaborative financial-modeling spreadsheet in three weeks, enabling finance teams to create live models from company accounting data using custom formulas, charts, templates, and version history
- Architected Google Sheets–style **real-time collaboration** using a revisioned operation log, optimistic concurrency, WebSocket synchronization, reconnect recovery, and conflict-safe version restoration

Shopify | Production Engineering Intern | Edge Infrastructure *September 2025 – December 2025*
- Moved **10+ services** onto a new SSL stack for merchant custom domains by building certificate issuance and renewal in **Ruby on Rails** on **Cloudflare**, **GCLB**, and **Google Certificate Manager**
- Raised monitoring coverage from **96.4% to 100%** with zero downtime during service migrations by deploying Cloudflare WAF rules through **Terraform**
- Resolved a **production outage** caused by Cloudflare API rate limiting by tracing the spikes with **Prometheus**, adding kill switches, and rewriting core retry logic with backoff; set up **Grafana** alerts to catch repeats
- Reduced false positives in edge bot detection by adding a **Lua** rule to the **Nginx** request pipeline

Shopify | Software Engineering Intern | Communications Platform Team *January 2025 – April 2025*
- Shipped features for a high-throughput communications platform using **React**, **Ruby on Rails**, and **GraphQL**, orchestrating **4M+ monthly** emails and push notifications across multiple Shopify services and teams
- Designed an approval workflow system for emails using a **polymorphic Rails concern** that empowers users in marketing automation, resulting in an **80% reduction** in faulty campaigns due to human error
- Cut non-actionable alerts by **70%** by adding filtering and categorization rules to **Bugsnag** error monitoring

Waterloo Aerial Robotics Group | Autonomy Lead *January 2024 – Present*
- Led a team of **15+ software developers** that contributed to back-to-back victories at AEAC 2024 and 2025
- Architected a robust **VSLAM pipeline for autonomous drone navigation,** integrating stereo-depth sensing and visual-inertial odometry (VIO) to ensure precise localization in GPS-denied environments
- Architected a fault-tolerant **ROS 2 behavior-tree mission controller** that sequences takeoff, timed waypoint laps, target reconnaissance, and landing while preserving progress through operator kill-switch interruptions
- Built and published a **Python** developer CLI using Git sparse checkout and dependency resolution to automatically configure, run, test, and diagnose individual robotics subsystems

Projects

varchar | *Rust, WebAssembly, Preact, SQL*
- Built a **SQL engine in Rust** that keeps a whole database in one string, with a hand-rolled lexer, parser, type checker, and a planner that makes a regex do an index scan's job
- **Fuzzed** the engine against a reference implementation, checking random CRUD sequences and predicates with proptest for identical results
- Ported the engine to **WebAssembly**, drawing each query's match spans onto the live database string so the planner's choices are visible as they run

salt | *Rust, TypeScript, React, Cloudflare Workers, KV*
- Built a CLI that reads local coding-agent session history, scanning **16 GB across 7,900 files in under 4 seconds** and leaving the scan bound by disk, not CPU
- Turned each local scan into a **shareable link**, serving reports and the static site from one Cloudflare Worker with KV records that expire themselves after 30 days
- Shipped **binaries for six platforms** from one npm package, cutting the install to 2 MB with no bundled runtime

cfa-lsp | *C++20, Cforall, LSP*
- Built a **language server for Cforall**, a research extension of C with polymorphism and return-type overloading, with 16 LSP features by forking the compiler to emit resolver output
- Made checks **2–4.6x faster** by profiling the compiler and skipping library bodies that took 94% of runtime
- Cut CI from **18 to 1.5 minutes** with cached toolchain and ccache builds, gating 220+ tests and every release

Technical Skills

Languages: Python, C, C++, Go, Ruby, Rust, TypeScript, JavaScript, SQL
Technologies: Ruby on Rails, React, React Native, Next.js, Django, GraphQL, Tailwind, Apache Spark
Nginx, Terraform, Git, Linux, AWS, GCP, WebAssembly, PyTorch, Pandas, Docker, Kubernetes